

Analysis

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Definition 1 (Compactness). A set $S \subset U$ (in a topological space) is called **compact** if for every open cover $\mathcal{T}$ of S there is a finite subcover of S .

Theorem 2. The image of a compact set under a continuous function is compact.

Proof idea: If $f : D \rightarrow \mathbb{R}$ is a continuous function and D is compact, then let $\mathcal{T}$ be any open cover of $f(D)$. Then taking the preimage of each $T \in \mathcal{T}$ gives an open cover of D . This has a finite subcover, $\mathcal{T}_0$. Then mapping these back to elements of $\mathcal{T}$ gives a finite subcover of $f(D)$.

Theorem 3. Let S be a compact set on $\mathbb{R}$ and $f : S \rightarrow \mathbb{R}$ be continuous. Then f achieves both its maximum and minimum on S .

Proof: By 2 theorem, $f(S)$ is compact. By Heine-Borel, it is closed and bounded and hence $\sup f(S) = \max f(S)$ and likewise for the minimum. $\square$

Theorem 4 (Rolle's Theorem). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function continuous on $[a, b]$ and differentiable on (a, b) such that $f(a) = f(b)$. Then there exists $c \in (a, b)$ s.t. $f'(c) = 0$.

Theorem 5 (Mean Value Theorem). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function that is continuous on $[a, b]$ and differentiable on (a, b) . Then there exists a $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof: Consider the function $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a)$. Then $g(a) = f(a)$ and $g(b) = f(a)$. Thus by Rolle's Theorem there exists a $c \in (a, b)$ s.t. $g'(c) = 0$ and the rest follows. $\square$

Theorem 6 (Taylor's Theorem). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function that has its first n derivatives continuous on $[a, b]$ and differentiable on (a, b) . Fix a point to expand around, x_0 . Let $x \in [a, b]$ with $x \neq x_0$. Then for any $n \in \mathbb{N}$ there exists and $c \in (x, x_0)$ such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

Idea of Proof:

- Construct a function $F(t)$ such that

1. F is continuous on (a, b) and differentiable on $[a, b]$.
2. $F(x) = f(x)$ and $F(x_0) = f(x_0)$
3. $F'(t) = 0$ forces the correct coefficient of $(x - x_0)^{n+1}$

- Use the mean value theorem on $F(t)$.

Noteworthy property of Modified Taylor Series: If $P(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!}(x - t)^k$ then

$$P'(t) = \frac{f^{(n+1)}(t)}{n!}(x - t)^n$$

Proof:(following Lay 28.2) Let $x \in [a, b]$ and fix $n \in \mathbb{N}$. Let

$$F(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!}(x - t)^k + M(x - t)^{n+1}$$

where M is the unique solution to

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + M(x - x_0)^{n+1}.$$

Note that by the noteworthy property mentioned above we have

$$F'(t) = \frac{f^{(n+1)}(t)}{n!}(x - t)^n + (n + 1)M(x - t)^n.$$

Note also that $F(x) = F(x_0) = f(x)$, so invoking the mean value theorem (or Rolle's Theorem) we obtain that there exists a c between x and x_0 strictly such that

$$F'(t) = 0 = \frac{f^{(n+1)}(t)}{n!}(x - t)^n + (n + 1)M(x - t)^n.$$

Solving for M gives $M = \frac{f^{(n+1)}(x_0)}{(n+1)!}$ as required. $\square$

Example 7. Let $f(x) = x^2 + 2x + 1$ on $[0, 3]$ with $x_0 = 0$. Let $n = 1$. Then there should exist a c such that

$$f(2) = 1 + 2(x - 0) + \frac{f^{(n+1)}(c)}{(n + 1)!}(x - 0)^{n+1}$$

$$1 + 2x + \frac{2}{2}x^2$$

so we recover our polynomial.

We now note that we can rephrase this theorem in terms of a remainder.

Theorem 8 (Taylor's Remainder Theorem). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $n + 1$ times differentiable and $x_0 \in \mathbb{R}$. Then if we approximate

$$f(x) \approx p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

then there exists a c between x and x_0 such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}.$$

Note though, that c depends on x .

Remark 9. Note that if f is a polynomial of degree k , then $p_n = f$ for all $n \geq k$ and all $x_0 \in \mathbb{R}$.

Lemma 10. Let f exist in a neighborhood of c and that $f''(c)$ exists. Then

$$\begin{aligned} f''(c) &= \lim_{h \rightarrow 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2} \\ f''(c) &= \lim_{h \rightarrow 0} \frac{(f(c+h) - f(c)) + (f(c-h) - f(c))}{h^2} \end{aligned}$$

Proof: Starting with the limit and using L'Hospital's rule we get:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2} &= \lim_{h \rightarrow 0} \frac{f'(c+h) - f'(c-h)}{2h} \\ &= \lim_{h \rightarrow 0} \frac{f'(c+h) - f'(c) + (f'(c) - f'(c-h))}{2h} \\ &= \frac{2f''(c)}{2} = f''(c) \square. \end{aligned}$$

Lemma 11. The following are equivalent definitions of the derivative.

- The usual one
- $f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$
- $f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c-h)}{2h}$